

VLAN Concepts Theory

Task 1 :- On a Cisco Packet Tracer switch, create two VLANs named 'Engineering' (VLAN 10) and 'Marketing' (VLAN 20) using the CLI commands.

ANS :-

1. **Open CLI :** Click the **Switch**, go to the **CLI** tab, and press **Enter**.

2. **Enter Config Mode :**

```
enable
configure terminal
```

3. **Create Engineering (VLAN 10) :**

```
vlan 10
name Engineering
```

4. **Create Marketing (VLAN 20) :**

```
vlan 20
name Marketing
end
```

5. **Verify :** Type show vlan brief to see both new VLANs in the list.

Task 2 :- Assign ports FastEthernet0/1 and FastEthernet0/2 to VLAN 10, and ports FastEthernet0/3 and FastEthernet0/4 to VLAN 20 on your switch.

ANS :-

1. Open Switch CLI and enter configuration mode :

```
enable
configure terminal
```

2. Assign Ports to VLAN 10 (Engineering) :

```
interface range fa0/1 - 2
switchport mode access
switchport access vlan 10
```

3. Assign Ports to VLAN 20 (Marketing) :

```
interface range fa0/3 - 4
switchport mode access
switchport access vlan 20
end
```

4. **Verify** : Type show vlan brief to check port assignments.

Task 3 :- Write a short document (in a text file) listing the VLAN IDs, names, and the ports assigned to each VLAN from your configuration.

ANS :-

VLAN CONFIGURATION REPORT

VLAN ID: 10

Name: Engineering

Ports: FastEthernet0/1, FastEthernet0/2

VLAN ID: 20

Name: Marketing

Ports: FastEthernet0/3, FastEthernet0/4

Task 4 :- Explain, in your own words, two benefits of using VLANs in a college campus network that has separate labs for coding and networking.

ANS :-

Better Security : Keeps coding and networking labs separate so students in one lab can't access or interfere with computers in the other.

Faster Network : Reduces network clutter (broadcast traffic) by locking communication inside each lab, making internet and lab speeds faster.